

Gennoor Tech

TRAIN. INNOVATE. BUILD.

GENNOOR ACADEMY · SELF-PACED ·
36 MIN

RAG Architectures — Foundations

A ~36-minute engineer's tour of retrieval-augmented generation. Embeddings, vector stores, hybrid search, re-ranking, evaluation, 6 failure modes, 5-component production architecture. Voice: Emma.

8

MODULES

36 min

DURATION

Intermediate

LEVEL

FREE

CERTIFICATE

Course Brief

DURATION	36 min · 8 modules · self-paced
FORMAT	Audio-narrated slide chapters with synchronized progress tracking. Interactive end-of-course Markdown builder you take to the next meeting. Free verifiable certificate on completion.
AUDIENCE	Anyone starting their AI journey — non-technical professionals, managers, analysts, L&D teams.
PREREQUISITES	No technical prerequisites. The material is written for business decision-makers and assumes only general business literacy.

Course Outcomes

- 1 When RAG fits — large/changing KB · citations + grounding · long-tail facts — and 3 cases when it's the wrong architecture
- 2 Embeddings as dense vectors — what they encode well (topic, paraphrase, semantic relations) and miss (IDs, negation, numerics, recency)
- 3 4 vector stores compared — pgvector, Pinecone, Azure AI Search, Weaviate — with the scale guide and cost traps to avoid
- 4 Hybrid search (BM25 + vector + semantic ranking) with RRF/weighted/cascade fusion — where retrieval quality actually lives
- 5 3 re-ranker types and the cost-quality curve — first 50 candidates lift 15-25%; beyond that, diminishing returns
- 6 Evaluation in two layers — retrieval (hit rate, MRR, recall@k) vs answer (RAGAS: groundedness, faithfulness, relevance) — never conflated
- 7 6 production failure modes (bad chunking, query intent mismatch, OOD queries, context overflow, metadata blind spots, stale knowledge) + targeted fixes
- 8 5-component production architecture + 4 trust trip-wires + interactive architecture-decision-record builder

Table of Content

8 modules · 36 min of audio-narrated learning · each module ends with a Monday-move you can apply that week.

MODULE 01

Why RAG and when not to use it

~5 MIN · NARRATED + INTERACTIVE

- When RAG fits (large/changing KB
 - citations
 - long-tail facts)
 - 3 cases when it's wrong
 - RAG vs long-context vs fine-tuning.
-

MODULE 02

Embeddings explained

~5 MIN · NARRATED + INTERACTIVE

- What they are
 - what they encode well (topic
 - paraphrase
 - semantic)
 - what they miss (IDs
 - negation
-

MODULE 03

Vector stores

~5 MIN · NARRATED + INTERACTIVE

- 4 stores compared (pgvector
 - Pinecone
 - Azure AI Search
 - Weaviate)
 - scale guide
 - 3 evaluation dimensions
-

MODULE 04**Hybrid search**

~5 MIN · NARRATED + INTERACTIVE

- Why pure vector fails
 - 3 fusion strategies (RRF
 - weighted
 - cascade)
 - semantic ranking as third layer
 - honest scoring.
-

MODULE 05**Re-ranking strategies**

~5 MIN · NARRATED + INTERACTIVE

- Why re-ranking
 - 3 types (cross-encoder
 - managed API
 - LLM)
 - cost-quality curve
 - re-rank a small top-N.
-

MODULE 06**Evaluation**

~5 MIN · NARRATED + INTERACTIVE

- 2 distinct layers (retrieval vs answer)
 - 3 retrieval metrics (hit
 - MRR
 - recall@k)
 - 3 answer metrics (RAGAS)
 - 3 eval-set options.
-

MODULE 07**Common RAG failures and fixes**

~5 MIN · NARRATED + INTERACTIVE

- 6 production failure modes (chunking
 - query intent
 - OOD
 - context overflow
 - metadata blind spots
 - stale knowledge) with targeted fixes.
-

MODULE 08**Capstone — production RAG**

~6 MIN · NARRATED + INTERACTIVE · CAPSTONE BUILDER

- The principle
- 5-component architecture
- 4 trust trip-wires
- Interactive · ADR builder

Taking This Further

The course is the starting point. The same content powers three concrete next steps you can engage Gennoor Tech to deliver inside your organisation:

4-week Pilot

- Fixed scope · fixed price · production artifact
- A working AI capability tied to the content of this course — set up on your data and tooling
- Suitable for: a single function or business unit ready to prove value

Org-wide Rollout

- 90-day adoption programme · executive + frontline tracks
- Champions network, prompt libraries, governance pack, weekly office hours
- Suitable for: 200-5,000-seat rollouts of Copilot, ChatGPT Enterprise, or similar

Continuous Build

- Quarterly retainer · embedded engineering + delivery + governance
- Multiple parallel AI initiatives sustained against an evolving roadmap
- Suitable for: large enterprises with portfolio-scale AI ambitions

Get in touch

Tell us about your AI ambitions, your team, and the constraints you are working under. We respond within one business day across India, GCC, SEA, US, and EU time zones.

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